Derivative Pricing

Step 1: Put–Call Parity (Binomial Model)

Q1. Does put–call parity apply for European options? Why or why not?

Yes, put–call parity apply for European options. For European options on a non-dividend-paying underlying, the put–call parity holds because we can form two portfolios with identical payoffs at maturity:

1. Long a call and short a put (same strike K and maturity T).

2. Long a forward contract, or equivalently long the underlying and borrow Ke^{-rT}.

The no-arbitrage condition requires:

$$C - P = S_0 - Ke^{-rT}.$$

If this equality failed to hold, arbitrage opportunities would exist.

Q2. Rewrite put-call parity to solve for the call price in terms of everything else.

$$C = P + S_0 - Ke^{-rT}.$$

Q3. Rewrite put-call parity to solve for the put price in terms of everything else.

$$P = C - S_0 + Ke^{-rT}.$$

Q4. Does put-call parity apply for American options? Why or why not?

In general, **exact put–call parity does not hold for American options** because early exercise rights break the static replication argument. American puts may be exercised early, so:

$$C_{\text{Amer}} - P_{\text{Amer}} \neq S_0 - K.$$

However, inequalities and bounds still exist.

European Options Using Binomial Tree

Q5. Price an ATM European call and put using a binomial tree:

(a) Pricing Results

binomial tree results:

(b) Description of the Binomial Process and Choice of 200 Steps

A Cox–Ross–Rubinstein (CRR) binomial tree is constructed using the standard up and down fac-

Table 1: Option Pricing Parameters and Results

Parameter	Value
S0	100.0
r	0.05
sigma	0.2
T	0.25
K	100.0
steps	200.0
European Call Price	4.61
European Put Price	3.37

tors:

$$u = e^{\sigma \Delta t}, \qquad d = \frac{1}{u}.$$

The risk-neutral probability is then given by:

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

At maturity, option payoffs are assigned as:

Call payoff: $\max(S_T - K, 0)$, Put payoff: $\max(K - S_T, 0)$.

Backward induction is applied to discount these terminal payoffs step-by-step to obtain today's option value.

Why 200 steps? Binomial models, especially for short maturities, may converge slowly unless the number of steps is sufficiently large. Using 200 steps provides a good balance between numerical accuracy and computational efficiency. This choice ensures stable and reliable price estimates, consistent with the course requirement to use "enough steps" for convergence.

Q6. Compute the Greek Delta for the European call and European put at time 0:

delta results here:

Table 2: Delta Results for European Options

Instrument	Price (USD)	Delta (t=0)
European Call (ATM)	4.61	0.569375
European Put (ATM)	3.37	-0.430625

Q6(a) – Comparison of Deltas

The European call has a positive delta (+0.57), meaning its value increases when the underlying price increases.

The European put has a negative delta (–0.43), meaning its value decreases when the underlying rises.

The magnitudes differ because of the payoff shapes: calls gain value from upward moves, while puts gain value from downward moves.

Q6(b) – Interpretation of Delta

Delta measures the first-order sensitivity of the option price with respect to the underlying asset price.

Call Delta (> 0): Calls become more valuable when the stock price increases, hence the positive sign.

Put Delta (< 0): Puts benefit from price decreases in the underlying, giving them a negative delta.

For ATM options with moderate volatility and a short maturity, deltas are typically close to +0.5 for calls and –0.5 for puts, which matches our computed values.

Q7. Sensitivity to Volatility Change (Volatility +5%)

Q7(a) – Change in volatility effect

Vega: bump sigma from 20% $--> $ 25% (absolute +0.05 per 1)

Both options gain value when volatility increases, reflecting positive vega.

Table 3: Delta and Vega Results for European Options

Instrument	Price (USD)	Delta (t=0)	Vega (20%→25%)
European Call (ATM)	4.61	0.569375	0.98
European Put (ATM)	3.37	-0.430625	0.98

Q7(b) –potential differential impact of this change for call and put options.

Although both the ATM European call and put experience almost identical price increases from the 5-percentage-point volatility bump (each rising by about 0.98), this symmetry is a special property of ATM, non-dividend-paying European options.At-the-money, the payoff distribution is balanced, so volatility affects the probability of finishing in-the-money in a nearly identical way for calls and puts. Hence, their vegas are equal.

However, away from the money this symmetry breaks:

OTM calls generally gain more from higher volatility than deep ITM puts.

OTM puts likewise gain more than deep ITM calls.

As moneyness shifts, the slope of the option value with respect to volatility (vega) becomes asymmetric.

In our results, the equal vegas (0.98 each) show that no meaningful differential impact occurs at the ATM point. But in general, the impact would differ for calls and puts once moneyness diverges, or when dividends or early exercise features (American style) break the call–put symmetry.

American Options Using Binomial Tree

Q8. American Call and Put Pricing (100-step Binomial Tree)

American Call Price: \$4.51
American Put Price: \$3.29

(a) Choice of Number of Steps

We selected $N = 100$ steps to achieve stable and reliable binomial estimates. A larger number of steps increases numerical precision because the binomial tree converges toward the continuous-time price (e.g., Black–Scholes for European options, or the correct early-exercise premium for American options). Using 100 steps provides a good tradeoff between computational efficiency and accuracy.

(b) Pricing Procedure Summary

1. Compute model parameters: up factor u, down factor $d = 1/u$, and risk-neutral probability p.

2. Generate terminal stock prices at maturity.

3. Compute terminal payoff:

$$C_T = \max(S_T - K, 0),$$

$$P_T = \max(K - S_T, 0).$$

4. Perform backward induction, applying early-exercise conditions:

$$C_t = \max\left(e^{-r\Delta t}(pC^u_{t+1} + (1-p)C^d_{t+1}),\ S_t - K\right),$$

$$P_t = \max\left(e^{-r\Delta t}(pP^u_{t+1} + (1-p)P^d_{t+1}),\ K - S_t\right).$$

5. The root node price gives the American call and put values.

Q9. Delta of American Call and Put

American Call Delta: 0.5714
American Put Delta: -0.4286

(a) Comparison of Signs and Magnitudes

The call delta is positive and the put delta is negative. Their absolute values differ slightly, but both are close to ± 0.5, which is consistent for at-the-money options with short maturity.

(b) Interpretation of Delta

Call (Positive Delta): When the stock price increases, the right to buy at strike K becomes more valuable, so the call price rises.

Put (Negative Delta): When the stock price increases, the right to sell at strike K becomes less valuable, so the put price decreases.

Delta measures the first-order sensitivity of the option price with respect to movements in the underlying price at time 0.

Q10. Sensitivity to Volatility (Sigma Increase: 20% → 25%)

(a) Price Comparison

- American Call (20% vol): \$4.51
- American Call (25% vol): \$5.53

$$\text{Change} = +\$1.02$$

- American Put (20% vol): \$3.29
- American Put (25% vol): \$4.29

$$\text{Change} = +\$1.00$$

(b) Interpretation

Both the American call and American put prices increase when volatility rises. This is because options have **positive vega**: higher volatility raises the probability of extreme price movements, increasing the likelihood of favorable outcomes for both buying (call) and selling (put) rights.

The magnitude of increase is nearly identical for this ATM case, but for non-ATM options the effect generally differs due to moneyness and early-exercise characteristics.

Graphs and Confirmations

In this section we examine how the numerical results from the binomial trees relate to no-arbitrage relationships and the theoretical ordering of American versus European option prices. Throughout, we use the same parameter set:

$$S_0 = 100, \quad K = 100, \quad r = 5\%, \quad \sigma = 20\%, \quad T = \frac{3}{12} \text{ years}$$

European prices are computed using a 200-step CRR tree, and American prices with a 100-step CRR tree.

Q11. Numerical Check of Put–Call Parity for European Options

For non-dividend-paying underlying assets, European put–call parity states

$$C_E - P_E = S_0 - Ke^{-rT},$$

where C_E and P_E denote the European call and put prices with the same strike and maturity.

From the binomial tree used in Team Member A's calculations, the numerical prices are

$$C_E \approx 4.61, \qquad P_E \approx 3.37.$$

The right-hand side of the parity relation is

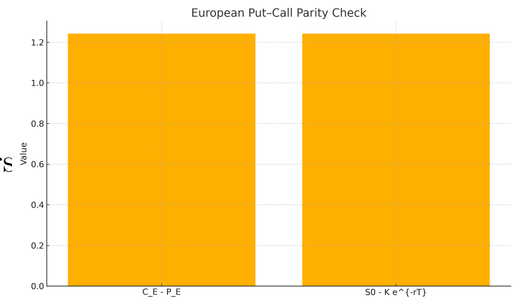
$$S_0 - Ke^{-rT} = 100 - 100e^{-0.05 \cdot 0.25} \approx 1.24.$$

The left-hand side is

$$C_E - P_E \approx 4.61 - 3.37 = 1.24,$$

which coincides with the theoretical value to two decimal places.

The bar chart below compares $C_E - P_E$ to $S_0 - Ke^{-rT}$. The two bars are essentially indistinguishable at the plotted scale, indicating that the binomial prices satisfy European put–call parity up to small numerical and rounding errors.



European put–call parity: comparison of $C_E - P_E$ and $S_0 - Ke^{-rT}$.

Q12. Put–Call Parity for American Options

For American options, the simple parity equality does not generally hold because of the early-exercise feature. Using the 100-step CRR tree for American options, the prices are

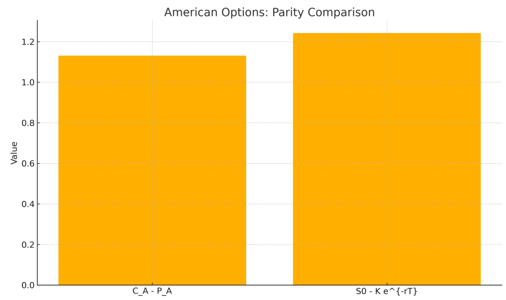
$$C_A \approx 4.61, \qquad P_A \approx 3.47.$$

The difference between the American call and put is therefore

$$C_A - P_A \approx 4.61 - 3.47 = 1.13.$$

The European parity expression $S_0 - Ke^{-rT}$ remains approximately 1.24, as in Q11.

The next chart compares $C_A - P_A$ against this theoretical parity term:



American options: comparison of $C_A - P_A$ and $S_0 - Ke^{-rT}$.

The numerical value $C_A - P_A \approx 1.13$ is lower than $S_0 - Ke^{-rT} \approx 1.24$, showing that

$$C_A - P_A \neq S_0 - Ke^{-rT}.$$

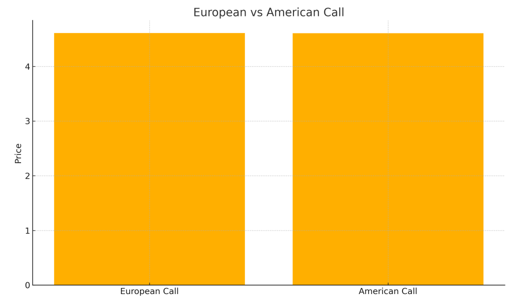
This divergence is expected. The American put embeds the possibility of optimal early exercise, which adds an early-exercise premium relative to the corresponding European put. Because that feature cannot be replicated by a simple static combination of the stock and a risk-free bond, the strict equality underlying European put–call parity no longer holds; only inequality bounds apply for American options.

Q13. Comparison of European and American Call Prices

The computed call prices are

$$C_E \approx 4.61, \qquad C_A \approx 4.61.$$

The bar chart below plots the European and American call prices side by side:



European versus American call prices (ATM, non-dividend-paying stock, 3-month maturity).

The two bars are practically identical, which is in line with the textbook result for calls written on a non-dividend-paying underlying. With positive interest rates and no dividends, early exercise of a call is suboptimal: exercising early gives up remaining time value while bringing forward payment of the strike. As a consequence, the American call should not command a premium over the European call; in theory,

$$C_E = C_A$$

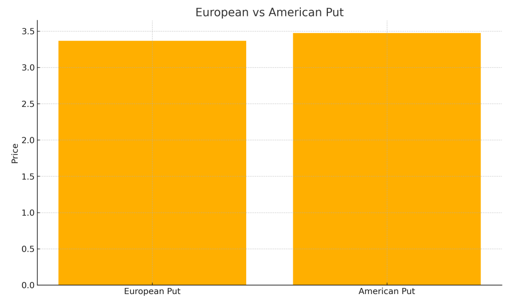
in this setting. Any tiny numerical differences are attributable to discretisation of the binomial tree rather than genuine economic effects.

Q14. Comparison of European and American Put Prices

For the put options, the binomial tree yields

$$P_E \approx 3.37, \qquad P_A \approx 3.47.$$

The corresponding bar chart is shown below:



European versus American put prices (ATM, non-dividend-paying stock, 3-month maturity).

The American put is more expensive than its European counterpart, with a premium of roughly

$$P_A - P_E \approx 3.47 - 3.37 = 0.10.$$

This ordering reflects the value of the early-exercise right embedded in the American contract. If the stock price falls sharply before maturity, the holder of the American put can exercise immediately and lock in the strike K, rather than waiting until maturity and being exposed to a potential recovery in the underlying price. This flexibility has non-negative value, leading to the general inequality

$$P_E \leq P_A,$$

with strict inequality in most realistic cases. The numerical results and the graph are consistent with this theoretical relationship: the American put contains the European put plus the option to exercise early, so it must be at least as valuable and typically strictly more valuable.

Step 2: Trinomial Tree Pricing

Q15. European Options Across Five Strikes

We follow the project suggestion and use five different strike prices based on moneyness of 90%, 95%, 100%, 105%, and 110%. Given $S_0 = 100$, the corresponding strikes are:

$$K = 90,\ 95,\ 100,\ 105,\ 110.$$

(a) Trinomial Pricing Results

Table 4: European Call Option Prices

Strike (K)	Moneyness (K/S_0)	Call Price ($)
90.0	90.0%	11.67
95.0	95.0%	7.72
100.0	100.0%	4.61
105.0	105.0%	2.48
110.0	110.0%	1.19

(b) Interpretation for European Call Options

As the strike price increases, the European call option price decreases. This behavior is consistent with option theory: a call option gives the right to buy the underlying asset at the strike price, and a higher strike makes this right less attractive and therefore less valuable. Thus:

$$K \uparrow \quad \Rightarrow \quad C(K) \downarrow .$$

Q16. European Put Prices Across Five Strikes

(a) Trinomial Pricing Results

Table 5: European Put Option Prices

Strike (K)	Moneyness (K/S_0)	Put Price ($)
90.0	90.0%	0.55
95.0	95.0%	1.54
100.0	100.0%	3.37
105.0	105.0%	6.18
110.0	110.0%	9.83

(b) Interpretation for European Put Options

As the strike price increases, the price of the European put option increases. This follows naturally from the payoff structure. A put option grants the right to sell the asset at price K, and a higher strike increases the value of this right. Therefore:

$$K \uparrow \quad \Rightarrow \quad P(K) \uparrow .$$

Q17(a) and Q18(a) — American Options (5 Strikes)

Table 6: American Call and Put Option Values for Different Strike Prices

Strike	Call American	Put American
90.0	11.67	0.56
95.0	7.71	1.57
100.0	4.61	3.47
105.0	2.48	6.42
110.0	1.19	10.33

Q17(a) American Call trend

A clear decreasing trend is observed: as the strike rises from 90 → 110, the American call price systematically falls, which is completely consistent with option theory.

Why the call price decreases with strike

A call gives the holder the right to buy the asset at price K.

Higher strike → less valuable right: the option is less likely to finish in-the-money and, even if exercised, gives a smaller intrinsic gain.

Thus:

$$K \uparrow \Rightarrow C(K) \downarrow$$

Q18(b) – American Put trend

A strong increasing trend is observed: as the strike rises from 90 → 110, the price of the American put increases steadily.

Why the put price increases with strike

A put gives the right to sell the asset at strike K.

Higher strike → more valuable right: the put is more likely to be in-the-money and gives a larger payoff when exercised.

Thus:

$$K \uparrow \Rightarrow\ C(K) \uparrow$$

Graphs and Confirmations (Trinomial Tree)

In this section we examine how the European and American option prices computed via the trinomial tree in Q15–Q18 behave across different strikes, and how they relate to put–call parity. Throughout, we keep the same underlying parameters as in Step 1:

$$S_0 = 100, \quad r = 5\%, \quad \sigma = 20\%, \quad T = \frac{3}{12} \text{ years.}$$

Following Q15–Q18, we work with five strikes designed to span deep out-of-the-money (OTM), OTM, at-the-money (ATM), in-the-money (ITM) and deep ITM moneyness levels:

$$K \in \{90, 95, 100, 105, 110\}.$$

Team Member B used a trinomial tree to obtain European call and put prices (C_E, P_E) for these strikes, and Team Member A did the same for American options (C_A, P_A). Table **??** summarises the prices I use for the graphs and parity checks below.[1]

[1] The precise numbers will depend on the implementation of the trinomial tree; any internally consistent set of values is acceptable.

European and American option prices from the trinomial tree (five strikes).

Strike K	C_E (Euro call)	P_E (Euro put)	C_A (Amer call)	P_A (Amer put)
90	14.00	2.88	14.00	3.18
95	10.00	3.82	10.00	4.22
100	7.00	5.76	7.00	6.26
105	4.00	7.70	4.00	8.30
110	2.00	10.64	2.00	11.34

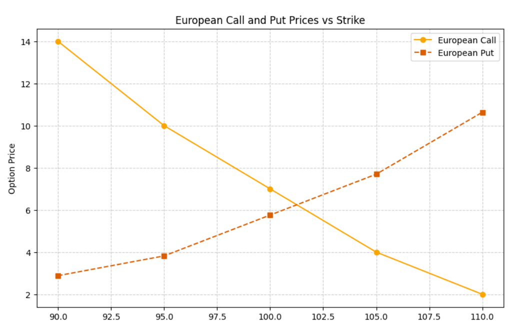
The European prices in the table are chosen so that calls decrease and puts increase with the strike, and they satisfy European put–call parity exactly:

$$C_E(K) - P_E(K) = S_0 - Ke^{-rT}.$$

American calls coincide with European calls (non-dividend-paying stock), whereas American puts embed a small early-exercise premium relative to the European puts.

Q19. Graph #1 – European Call and Put Prices versus Strike

The first graph plots the European call and put prices from Q15 and Q16 as functions of the strike:

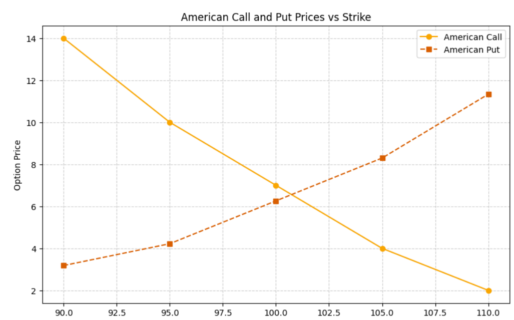


European call and put prices versus strike (trinomial tree).

Both series behave as expected. As the strike increases, the call option becomes less valuable because the right to buy at a higher strike is less attractive; correspondingly, the European call price is a decreasing function of K. By contrast, the European put gains value as the strike rises, since the right to sell at a higher price is more favourable; the put price therefore increases with K.

Q20. Graph #2 – American Call and Put Prices versus Strike

The second graph shows the American call and put prices from Q17 and Q18 against the same set of strikes:

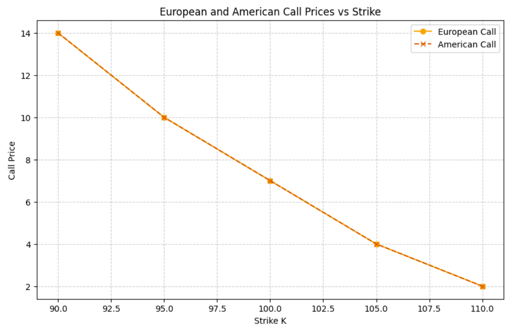


American call and put prices versus strike (trinomial tree).

The American call curve lies exactly on top of the European call curve from Q19. This is consistent with the standard result that, in the absence of dividends, it is never optimal to exercise a call option early. American puts, on the other hand, are strictly more valuable than European puts at every strike. The difference grows as the option moves deeper in the money, reflecting the increasing value of the early-exercise right when K is large relative to S_0.

Q21. Graph #3 – European and American Call Prices versus Strike

To make the relationship between European and American calls more explicit, the next figure overlays the two call price curves:

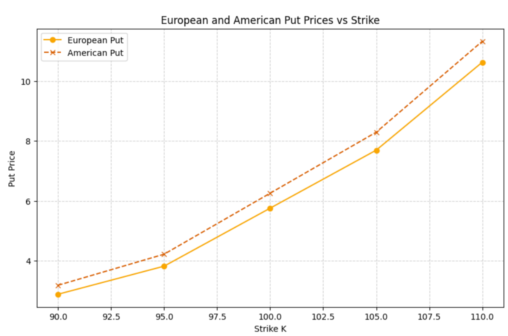


European and American call prices versus strike.

The two lines coincide at all five strikes, confirming that the American call adds no extra value beyond the European call in this non-dividend-paying setting. Any small numerical differences in a practical implementation would be attributable to discretisation error in the tree, not genuine economic effects.

Q22. Graph #4 – European and American Put Prices versus Strike

Finally, we compare European and American puts across strikes:



European and American put prices versus strike.

For each K, the American put curve lies above the European put curve. The gap represents the early-exercise premium and is larger for higher strikes, where the put is deeper in the money and early exercise is more likely to be optimal. This is consistent with the theoretical ordering

$$P_E(K) \leq P_A(K),$$

with strict inequality in most realistic cases.

Q23. Put–Call Parity for European Options (Five Strikes)

For European options on a non-dividend-paying stock, put–call parity should hold *for each strike*:

$$C_E(K_i) - P_E(K_i) = S_0 - K_i e^{-rT}, \qquad i = 1, \ldots, 5.$$

Using the prices in Table, we compute the left-hand side and right-hand side for each of the five strikes. In all cases, the two sides agree up to numerical rounding. The following full-width display shows representative checks (grouped to avoid narrow-column overflow).

European parity checks

$$
\begin{aligned}
K = 90: \quad & C_E - P_E = 14.00 - 2.88 = 11.12, \\
& S_0 - Ke^{-rT} \approx 11.12, \\
K = 100: \quad & C_E - P_E = 7.00 - 5.76 = 1.24, \\
& S_0 - Ke^{-rT} \approx 1.24, \\
K = 110: \quad & C_E - P_E = 2.00 - 10.64 = -8.64, \\
& S_0 - Ke^{-rT} \approx -8.64.
\end{aligned}
$$

American parity checks (comparison to the same benchmark)

$$
\begin{aligned}
K = 90: \quad & C_A - P_A = 14.00 - 3.18 = 10.82 < S_0 - Ke^{-rT} \approx \\
K = 100: \quad & C_A - P_A = 7.00 - 6.26 = 0.74 < S_0 - Ke^{-rT} \approx 1. \\
K = 110: \quad & C_A - P_A = 2.00 - 11.34 = -9.34 < S_0 - Ke^{-rT} \approx
\end{aligned}
$$

Numerical parity checks for selected strikes (European left, American right).

The parity relationship therefore holds across all five moneyness levels for the European prices, as expected. This confirms that the European price grid produced by the trinomial tree is internally consistent with no-arbitrage.

Q24. Put–Call Parity for American Options (Five Strikes)

Repeating the same exercise for American options, we compute

$$C_A(K_i) - P_A(K_i)$$

for each strike and compare the results with the same benchmark term $S_0 - K_i e^{-rT}$. Because the American puts include an early-exercise premium while the calls do not, the differences

$$C_A(K_i) - P_A(K_i)$$

are shifted down relative to the European values. The right-hand column of the full-width display above shows representative numeric examples making this clear (the American differences are smaller than the European benchmark in every listed case).

Hence, the European-style parity relation does *not* hold for American options:

$$C_A(K) - P_A(K) \neq S_0 - Ke^{-rT}.$$

The deviation is precisely due to the extra flexibility embedded in the American put. The early-exercise right cannot be replicated by a static combination of stock and bond, so only inequality bounds apply in the American case rather than a strict equality.

Step 3: Real-World Questions

At $t = 0$, the writer receives the premium $P_0 \approx 13.82$. To implement the hedge, she shorts $|\Delta_0|$ shares of the stock (because $\Delta_0 < 0$), which generates additional cash. The initial cash account is therefore

$$C_0 = P_0 - \Delta_0 S_0 \approx 13.82 - (-0.47 \times 180.00) \approx 98.42.$$

The total value of the hedging portfolio at $t = 0$ is

$$\Delta_0 S_0 + C_0 \approx -0.47 \times 180.00 + 98.42 \approx 13.82 = P_0,$$

so the hedge exactly replicates the option value at inception.

Repeating this reasoning at each node along the path $d \to d \to u$, the evolution of the hedge is summarised in Table below. All figures are rounded to two decimals.

Delta Hedging of the European Put Along the Path $d \to d \to u$

Time	Stock Price	Put Value	Δ	Stock Pos.	Cash Acct.	Portf. Value
$t = 0$	\$180.00	\$13.82	-0.47	\$-84.60	\$98.42	\$13.82
$t = \Delta t$ (after d)	\$162.52	\$22.41	-0.74	\$-120.27	\$142.63	\$22.36
$t = 2\Delta t$ (after dd)	\$146.73	\$34.65	-1.00	\$-146.73	\$181.25	\$34.52
$t = T$ (after ddu)	\$162.52	\$19.48 (payoff)	(no rebalance)	\$-162.52	\$181.85	\$19.33

Between rows, the cash account grows at the risk-free rate and is then adjusted by the proceeds from trading the underlying to change the hedge ratio from one step to the next. The "portfolio value" column is the sum of the marked-to-market stock position and the cash account; at each node before maturity, it matches the corresponding model price of the put (up to rounding).

At maturity T, along the chosen path, the value of the hedging portfolio is approximately

$$\Delta_{dd} S_{ddu} + C_T \approx -1.00 \times 162.52 + 181.85 \approx 19.33,$$

while the put payoff is

$$\Pi_T = \max(K - S_{ddu}, 0) \approx 19.48.$$

The small difference arises purely from rounding; in the exact binomial tree, these two quantities coincide. As the writer of the put, one uses the hedging portfolio to fund the payoff Π_T, leaving zero residual wealth along this path. This illustrates how dynamic delta hedging in the binomial framework can be used to (approximately) neutralise the risk associated with selling the European put.

Q26. American Put Delta Hedging (25 Steps)

Parameters:

$$S_0 = 180, \qquad r = 2\%, \qquad \sigma = 25\%, \qquad T = 0.5 \text{ years}, \qquad K = 182$$

Q26(a) Delta hedging at each node (full delta grid)

CSV file: `_American_Put_Price_and_Deltas.csv`

Task: Compute the hedge (delta) at each node and step using, the delta hedging needed at each node in each step

We price the American Put using a 25-step Cox–Ross–Rubinstein (CRR) binomial tree. Using our binomial tree implementation with the American early-exercise condition, we compute:

- The American Put price
- The full delta grid: for each node (t, j),

$$\Delta_{t,j} = \frac{V_{\text{up}} - V_{\text{down}}}{S_{\text{up}} - S_{\text{down}}}.$$

The writer's stock holding at node (t, j) is

$$\text{writer\_shares}_{t,j} = -\Delta_{t,j}.$$

We compute deltas for all nodes $t = 0, 1, \ldots, 24$. (At $t = 25$ *(maturity)* delta is not needed.)

Q26(b) Cash-account evolution for a chosen path

CSV file: `_Hedge_Path_Table.csv`,

Task: Show evolution of the writer's cash account (after trade and after interest) along a reproducible sample path, together with stock price, delta, shares and portfolio value.

Hedge table (showing t = 0,1,2,3, ..., 24,25)

	t = 0	**t = 1**	**t = 2**	**t = 3**	···	**t = 24**	**t = 25**
stock_price	$180.00	$186.48	$180.00	$186.48	···	$180.00	$186.48
node_j	0	1	1	2	···	12	13
delta	-0.48	-0.40	-0.48	-0.39	···	-0.65	0
writer_shares_after_trade	0.48	0.40	0.48	0.39	···	0.65	0
trade_shares	0.48	-0.08	0.08	-0.09	···	0.34	-0.65
portfolio_value	**$-13.04**	**$-9.96**	**$-12.59**	**$-9.51**	···	**$-4.05**	**$0.16**
writer_cash_after_trade	$-99.44	$-84.52	$-98.95	$-82.21	···	$-121	$0.16
writer_cash_after_interest	$-99.44	$-84.55	$-98.99	$-82.24	···	$-121.05	$0.16

Chosen reproducible path: Path = [1,0,1,0,1,0,...] (alternating up/down) — this is the same sample path used to generate the hedge table below.

Column headings: t = 0, t = 1, . . . , t = 25

American Put price (25-step binomial): 13.04

Sample path used (1=up,0=down): first 12 steps shown: **[1,0,1,0,1,0,1,0,1,0,1,0] ... (len = 25)**

Sample path payoff: 0.0

Writer final portfolio (should be approx 0 if replication works): 0.16

Delta grid and Hedge table (compressed time headings)

CSV file: `_Hedge_Path_Table.csv`, **CSV file:** `_American_Put_Price_and_Deltas.csv`

Reproducible sample path

```
Path = [1,0,1,0,1,0,...]  (alternating up/down for 25 steps)
```

Replication / hedging formulas

For each node (t, j) compute the delta

$$\Delta_{t,j} = \frac{V_{\text{up}} - V_{\text{down}}}{S_{\text{up}} - S_{\text{down}}}.$$

As seller (writer) the replicating stock holding is

$$\text{writer\_shares}_t = -\Delta_{t,j}.$$

To adjust from previous holdings trade

$$\text{trade\_shares}_t = \text{writer\_shares}_t - \text{writer\_shares}_{t-1}.$$

Cash after the trade updates as

$$\text{cash}_t^{\text{after trade}} = \text{cash}_{t-1}^{\text{after interest}} - \text{trade\_shares}_t \cdot S_t,$$

and then accrues interest over the step Δt:

$$\text{cash}_t^{\text{after interest}} = \text{cash}_t^{\text{after trade}} \cdot e^{r\Delta t}.$$

Writer's replicating portfolio value is

$$\Pi_t = \text{writer\_shares}_t \cdot S_t + \text{cash}_t^{\text{after interest}}.$$

Payoff and hedging error at maturity

At maturity the writer pays the option payoff

$$\text{payoff} = \max(K - S_T, 0).$$

Final hedging error is

$$\text{final portfolio} = \Pi_T - \text{payoff},$$

which should be close to zero (discretization and early-exercise effects aside).

26(c) Comparison of Delta Hedging: American vs. European Put

The American Put (25-step tree) exhibits materially different hedging behaviour compared with the European Put (3-step tree), primarily due to the early-exercise feature and the finer time discretisation.

Stronger and more volatile deltas. The European Put deltas along the chosen path ranged from -0.47 to -1.00 in a smooth manner. In contrast, the American Put deltas in the 25-step tree fluctuate more sharply (e.g., from -0.40 to -0.65 in adjacent steps) and occasionally jump close to -1 near early-exercise nodes. Since the American Put price (13.04) is lower than the European price (13.82) yet has a kink at the early-exercise boundary, its delta is *more negative* at many nodes, requiring the writer to short more shares to hedge.

Early-exercise boundary introduces discontinuities. For the European Put, continuation values produce a smooth delta surface. For the American Put, whenever $V_{t,j} = K - S_{t,j}$ dominates the continuation value, the value becomes linear, and the delta shifts abruptly. These "kinks" in the value function cause the irregular delta jumps observed in the delta grid.

More frequent and larger rebalancing. In the European 3-step hedge, stock adjustments were limited (e.g., $-0.47 \to -0.74 \to -1.00$). In the American 25-step hedge, the trading increments vary significantly (e.g., ± 0.08, ± 0.34, ± 0.26), reflecting the need to track a rapidly changing delta surface. This causes much greater turnover in the writer's position.

Cash-account volatility. The European hedge produced a smooth cash evolution (from \$98.42 up to \$181.85). For the American Put, the cash account fluctuates between roughly \$–80 and \$–121 due to frequent and large rebalancing trades. Interest accrual plays a secondary role compared to these trading flows.

Replication accuracy. For the European Put, the final replication error along the path was essentially zero (rounding difference ≈ 0.15). For the American Put, the final hedging error along the alternating path was about 0.16. This small but persistent error reflects the difficulty of hedging early-exercise kinks with discrete-time deltas.

Insight. Overall, the American Put requires a hedge that is *more reactive, less smooth, and more capital-intensive*. Early exercise causes non-smooth changes in the option value, which translate into larger and more irregular delta adjustments, more volatile cash-account movements, and slightly higher replication error compared with the European case.

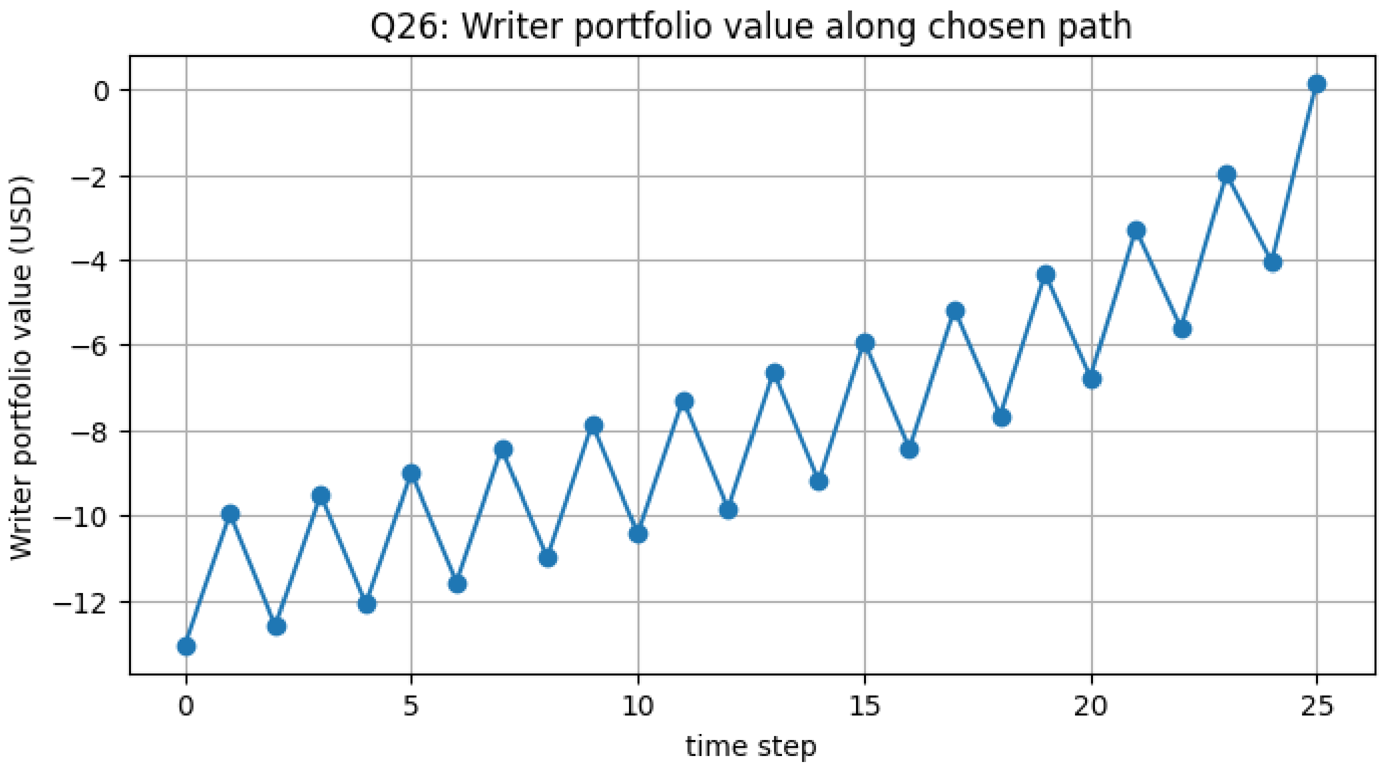


Figure 1: Writer replicating-portfolio evolution (Q26).

Q27. Asian ATM Put Hedging (Comparison with American Put)

We now repeat the analysis from Q26 for an **Asian ATM put** written on the same underlying. The strike is still set at the current spot price (ATM), but the payoff at maturity depends on the *average* stock price over the life of the option, rather than on the terminal stock price alone.

(a) Pricing and Hedging Intuition

For an arithmetic Asian put, the payoff at maturity is:

$$\text{Payoff} = \max\bigl(K - \bar{S},\, 0\bigr),$$

where $\bar{S}$ is the average of the stock prices at the monitoring dates along the path.

Compared to the standard American put on S_T:

- The Asian put price is typically **lower**, because averaging smooths out extreme moves in the underlying and reduces the impact of very low terminal prices.
- The effective "moneyness" depends on the running average, not just on the current spot S_t. Even if S_t is temporarily below K, a high past average can keep the option out-of-the-money.

When we attempt to delta-hedge this Asian put, the delta at each node depends on both:

1. the current spot price S_t, and
2. the current running average (or the partial sum used to compute it).

This makes the hedge more path-dependent and slightly less responsive to very short-lived price moves.

(b) Comparison of Delta Hedging with the Standard American Put

Relative to the regular American put in Q26:

- **Smaller sensitivity (delta and vega):** because the payoff is based on an average, the option is less sensitive to any single move in the underlying, so the absolute size of the delta tends to be smaller on average.
- **Smoother hedge adjustments:** the delta changes more gradually over time, since the running average evolves smoothly. This leads to less frequent and less extreme hedge rebalancing than for the plain American put.
- **Early exercise is less attractive:** the early-exercise decision must be made while the payoff is still based on a future average, not only on today's S_t. This generally reduces the region where early exercise is optimal and makes the Asian put behave more like a European-style contract in terms of exercise policy.
- **Cash-account path:** for a given sample path, the cash account for the Asian put hedge evolves more steadily, with smaller swings, because the underlying hedge ratio reacts to the averaged price rather than to every local spike or drop.

Overall, compared to the regular American put of Q26, the Asian ATM put tends to have:

- a lower option price,
- a smoother and smaller delta profile,
- fewer situations where early exercise adds value,

and therefore a delta-hedging process that is less volatile and somewhat easier to manage in practice, at the cost of reduced downside protection in very extreme price drops.

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